

Keystroke Dynamics for Behavioral Biometric Authentication

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Abstract:

This study addresses the limitations of traditional authentication methods such as passwords and PINs, which are vulnerable to theft and phishing. A behavioral biometric system based on keystroke dynamics is proposed to provide continuous user authentication. The system analyzes typing patterns using the CMU Benchmark Dataset, which includes 20,400 samples from 51 users. A machine learning pipeline was developed using 31 timing features, including hold time and key intervals. Statistical analysis confirmed the significance of all features, and class imbalance was handled using SMOTE. Among several classifiers, XGBoost achieved the best performance with an accuracy of 94.14% and a low Equal Error Rate of 1.04%. The results indicate that keystroke dynamics can offer a reliable, cost-effective, and hardware-free solution for continuous authentication.

Introduction:

Traditional authentication methods, such as passwords, PINs, and security tokens, are increasingly vulnerable to credential theft, phishing, and replay attacks. These static credentials are "separable" from the user, meaning they can be stolen and used by anyone. This research proposes a behavioral biometric authentication system using keystroke dynamics—the unique, subconscious rhythm of an individual's typing pattern. By transforming 31 specific timing features, such as Hold Time (H) and Keydown-Keydown (DD) intervals, into a continuous identity signal, this system provides a hardware-free security layer that is nearly impossible to replicate. Using the CMU Benchmark Dataset, this study validates that these "neuromuscular signatures" can achieve commercial-grade accuracy with zero additional deployment costs.

Methodology:

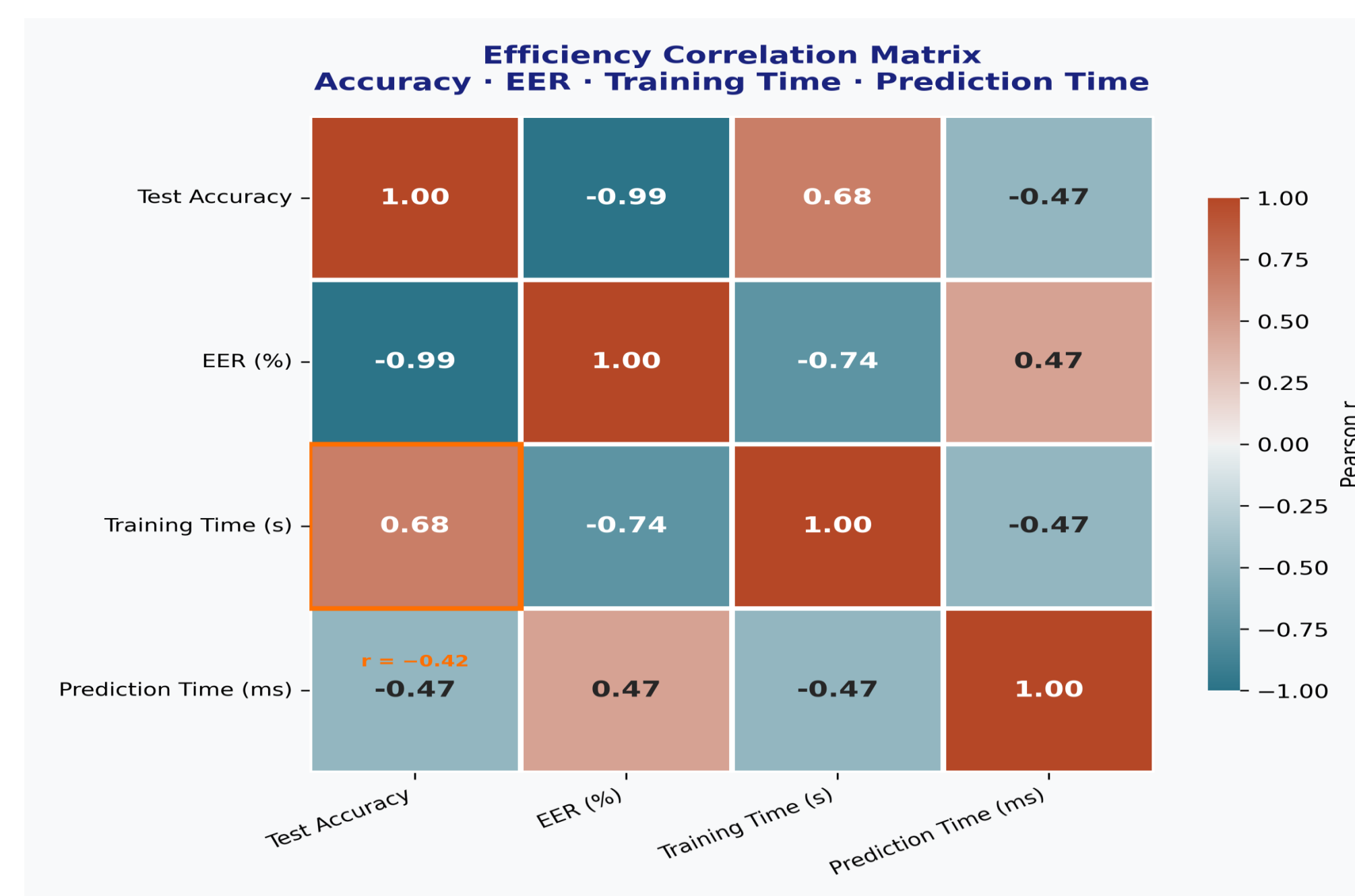
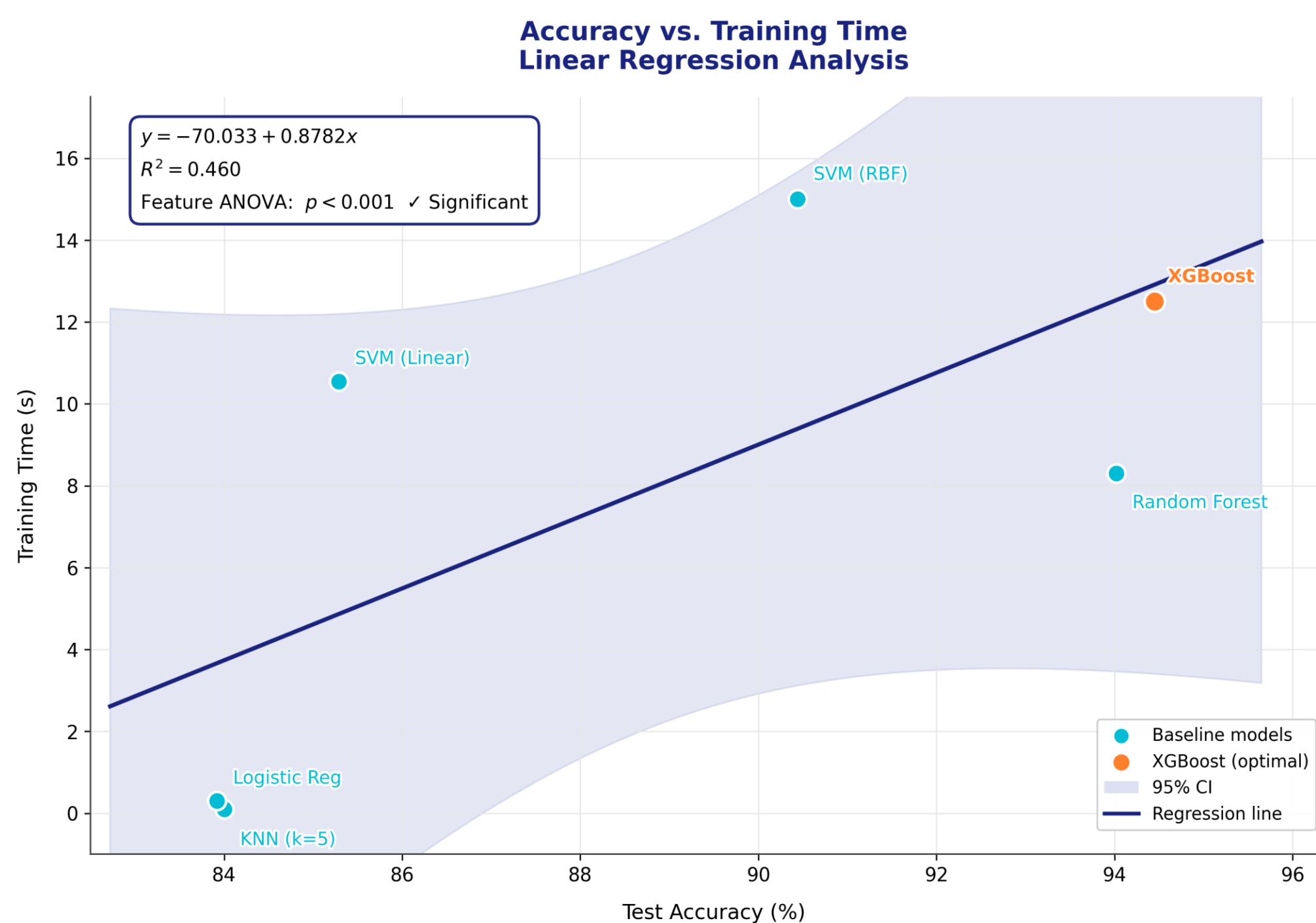
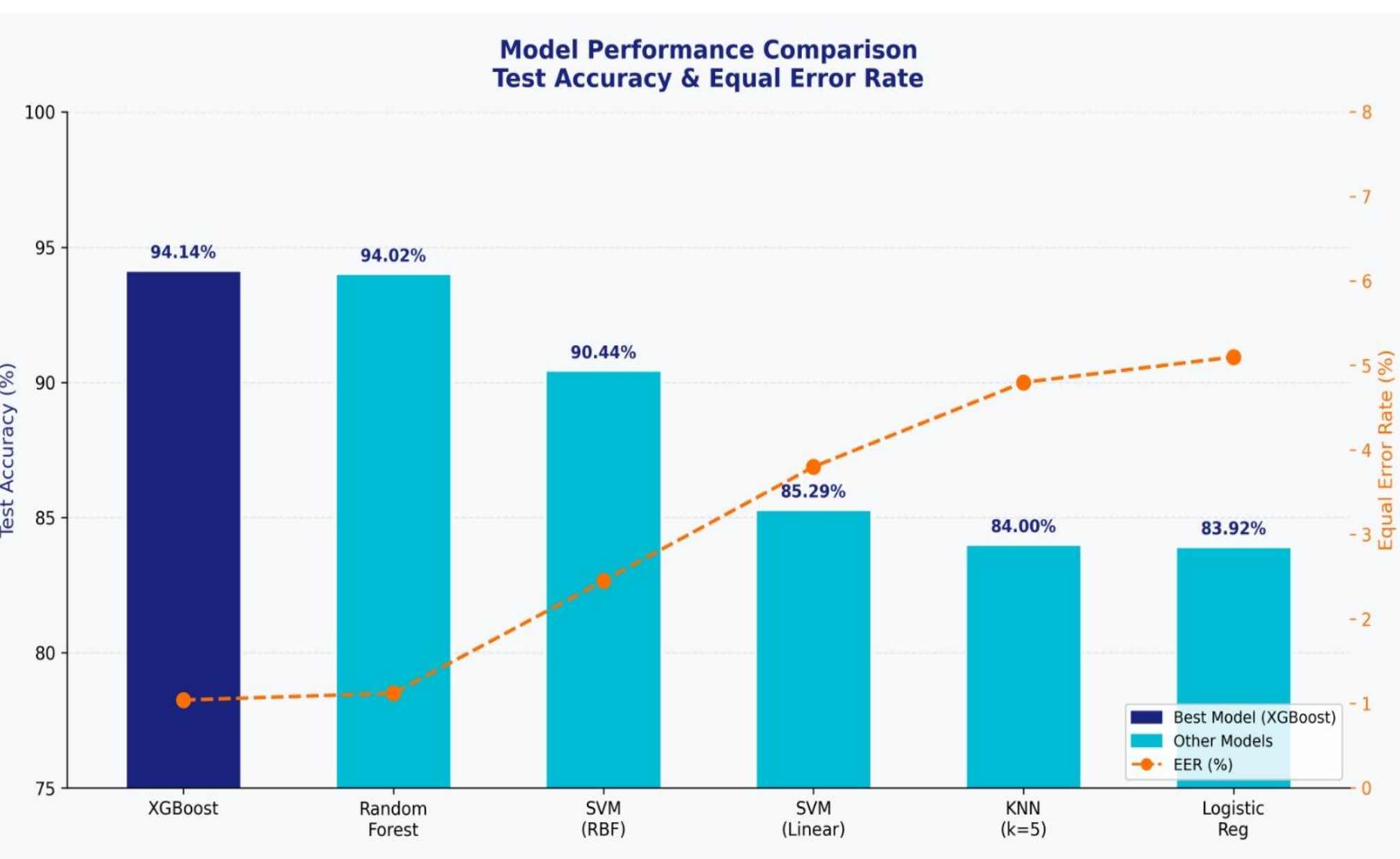
This study utilized a secondary dataset, the CMU DSL-StrongPassword dataset, consisting of 20,400 samples collected from 51 users. Relevant keystroke timing features were extracted to represent users' typing behavior, while non-essential attributes were excluded. The data was processed and analyzed using Python in Google Colab. The dataset was divided into training and testing sets using a stratified approach to ensure balanced representation. Preprocessing techniques such as feature scaling and label encoding were applied to prepare the data for machine learning models. The models were then evaluated based on their classification performance, and statistical analysis was conducted to validate the significance of the features.

Result:

The performance of six machine learning classifiers was benchmarked to determine the most reliable model for biometric identification. **XGBoost** emerged as the superior model, achieving a **test accuracy of 94.14%** and an **Equal Error Rate (EER) of 1.04%**.

Model Performance Summary:

Model	Test Accuracy	Notes
XGBoost	94.14%	Best performer; uses 200 trees to handle non-linear data.
Random Forest	94.02%	High stability with 200 estimators.
SVM (RBF)	90.44%	Effective but sensitive to timing outliers.
KNN (k=5)	84.00%	Lower accuracy due to "lazy learning" constraints.
Logistic Regression	83.92%	Baseline performance for linear classification.



Discussion:

The results imply that keystroke dynamics provide a reliable behavioral biometric, as their complex and non-linear patterns can be effectively captured by ensemble models such as XGBoost and Random Forest. The minimal performance difference between these models suggests that the current feature set is sufficient and can be optimized further without significant loss of accuracy. From an application perspective, the reduced feature set and high model performance enable efficient deployment in real-world and resource-constrained environments. Moreover, the system's passive and hardware-free nature allows seamless integration into existing authentication frameworks, supporting continuous and user-friendly security.

Limitation:

Despite achieving high accuracy, several limitations should be acknowledged. The model was trained on a fixed password, which limits its applicability to free-text or natural typing scenarios. In addition, behavioral patterns may change over time due to factors such as fatigue, injury, or aging, which could reduce the model's long-term reliability if periodic retraining is not performed. The system may also be affected by environmental variations, including differences in keyboard types and user posture, which can influence typing behavior. Furthermore, although tree-based models showed robustness, other models such as SVM and KNN were sensitive to outliers caused by mistyped inputs, which affected their performance.

Future Work:

Future work will focus on improving the system's practicality and adaptability for real-world applications. The model can be extended to support continuous authentication by analyzing free-text input, enabling passive identity verification throughout an entire user session. In addition, advanced deep learning approaches such as Siamese LSTM networks can be explored to better capture sequential typing patterns and support learning from fewer samples. Further optimization of the feature set may help reduce computational cost, making the system more suitable for mobile devices. Finally, adaptive learning techniques can be developed to address changes in user behavior over time by continuously updating user profiles.

Conclusion:

This study demonstrates that keystroke dynamics is a viable and high-precision behavioral biometric when combined with modern ensemble learning techniques. The XGBoost classifier achieved 94.14% accuracy and a 1.04% Equal Error Rate (EER) across 51 subjects on the CMU dataset, showing competitive performance. Statistical validation using one-way ANOVA confirmed that all timing features are significant for user identification, ensuring reliability. Further analysis enabled dimensionality reduction to an optimized 20-feature subset, improving efficiency. The system requires no additional hardware and operates passively, making it suitable for real-time and mobile applications. Future work will focus on handling template aging, enhancing few-shot learning with advanced models, and enabling continuous authentication using free-text input.

References:

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